

$$a) \quad f(t) = 1 + 2t - 4t^4$$

Since the Laplace transform is linear, we can find the transform of each term

$$f(t) = t^n \Rightarrow F(s) = \frac{n!}{s^{n+1}}$$

$$\mathcal{L}(1) = \frac{1}{s}$$

$$2\mathcal{L}(t) = \frac{2}{s^2}$$

$$4\mathcal{L}(t^4) = \frac{96}{s^5}$$

$$\underline{F(s) = \frac{1}{s} + \frac{2}{s^2} - \frac{96}{s^5}}$$

$$b) \quad f(t) = 2t^2 e^{3t}$$

$$f(t) = t^n e^{at} \Rightarrow F(s) = \frac{n!}{(s-a)^{n+1}}$$

$$\mathcal{L}(f(t)) = 2 \frac{2!}{(s-3)^3}$$

$$\underline{F(s) = \frac{4}{(s-3)^3}}$$

$$c) \quad f(t) = e^{-2t} \sin(5t)$$

$$f(t) = e^{at} \sin(bt) \Rightarrow F(s) = \frac{b}{(s-a)^2 + b^2}$$

$$\mathcal{L}(f(t)) = \frac{5}{(s-(-2))^2 + 5^2}$$

$$\underline{F(s) = \frac{5}{(s+2)^2 + 25}}$$

Prob. 2 - Inverse Laplace transform

$$a) \quad F(s) = \frac{3}{s} + \frac{1}{s^8}$$

Since the Laplace transform is linear, we can find the transform of each term

$$F(s) = \frac{1}{s} \Rightarrow f(t) = 1, \quad F(s) = \frac{n!}{s^{n+1}} \Rightarrow f(t) = t^n$$

$$3\mathcal{L}^{-1}\left(\frac{1}{s}\right) = 3 \cdot 1 = 3$$

$$\mathcal{L}^{-1}\left(\frac{1}{s^8}\right) = n+1=8 \Rightarrow n=7, \quad \frac{7!}{s^8} \Rightarrow \frac{1}{7!} t^7$$

$$\underline{f(t) = 3 + \frac{1}{7!} t^7}$$

$$b) \quad F(s) = \frac{2s+1}{s^2-s-2} = \frac{2s}{(s+1)(s-2)} + \frac{1}{(s+1)(s-2)}$$

Partial fraction decomposition

$$\frac{2s}{(s+1)(s-2)} = \frac{A}{(s+1)} + \frac{B}{(s-2)} \Rightarrow 2s = s(A+B) - 2A+B \Rightarrow \begin{cases} A+B=2 \\ -2A+B=0 \end{cases} \Rightarrow A = \frac{2}{3}, B = \frac{4}{3}$$

$$\frac{1}{(s+1)(s-2)} = \frac{A}{(s+1)} + \frac{B}{(s-2)} \Rightarrow 1 = s(A+B) - 2A+B \Rightarrow \begin{cases} A+B=0 \\ -2A+B=1 \end{cases} \Rightarrow A = -\frac{1}{3}, B = \frac{1}{3}$$

$$F(s) = \overset{1.}{\frac{2}{3(s+1)}} + \overset{2.}{\frac{4}{3(s-2)}} - \overset{3.}{\frac{1}{3(s+1)}} + \overset{4.}{\frac{1}{3(s-2)}}$$

$$F(s) = \frac{1}{(s-2)} \Rightarrow f(t) = e^{2t}$$

$$\mathcal{L}^{-1}(F_1(s)) = \frac{2}{3} e^{-t}$$

$$\mathcal{L}^{-1}(F_2(s)) = \frac{4}{3} e^{2t}$$

$$\mathcal{L}^{-1}(F_3(s)) = \frac{1}{3} e^{-t}$$

$$\mathcal{L}^{-1}(F_4(s)) = \frac{1}{3} e^{2t}$$

$$f(t) = \frac{2}{3} e^{-t} + \frac{4}{3} e^{2t} - \frac{1}{3} e^{-t} + \frac{1}{3} e^{2t}$$

$$c) \quad F(s) = \frac{1}{(s+1)(s^2+1)}$$

Partial fraction decomposition

$$\frac{1}{(s+1)(s^2+1)} = \frac{A}{(s+1)} + \frac{Bs+C}{(s^2+1)} \Rightarrow 1 = A(s^2+1) + (Bs+C)(s+1) \Rightarrow \begin{cases} A+B=0 & (s^2) \\ B+C=0 & (s) \\ A+C=1 & (1) \end{cases} \Rightarrow A = \frac{1}{2}, B = -\frac{1}{2}, C = \frac{1}{2}$$

$$F(s) = \frac{1}{2(s+1)} + \frac{\frac{1}{2}}{2(s^2+1)} - \frac{\frac{1}{2}}{2(s^2+1)}$$

$$F(s) = \frac{1}{(s-2)} \Rightarrow f(t) = e^{2t}, F(s) = \frac{a}{(s^2+a^2)} \Rightarrow f(t) = \sin(at), F(s) = \frac{s}{(s^2+a^2)} \Rightarrow f(t) = \cos(at)$$

$$\mathcal{L}^{-1}(F_1(s)) = \frac{1}{2} e^{-t}$$

$$\mathcal{L}^{-1}(F_2(s)) = \frac{1}{2} \sin(t)$$

$$\mathcal{L}^{-1}(F_3(s)) = \frac{1}{2} \cos(t)$$

$$f(t) = \frac{1}{2} (e^{-t} + \sin(t) - \cos(t))$$

Prob 3 - True or False

a) If f and g are two functions for which the Laplace transform exists, then

$$\mathcal{L}(f+g) = \mathcal{L}(f) + \mathcal{L}(g) \Rightarrow \underline{\text{True because of linearity}}$$

- b) If f and g are two functions for which the Laplace transform exists, then
 $\mathcal{L}(f \cdot g) = \mathcal{L}(f) \cdot \mathcal{L}(g) \Rightarrow$ kind of true because of convolution
- c) If the function f satisfies $0 \leq f(t)$, $t \geq 0$, then $\mathcal{L}(f)(s) \geq 0$ for all s for which $\mathcal{L}(f)(s)$ exists. $\Rightarrow$ True because of the def. of $\mathcal{L}$. $f(t)$ pos $\Rightarrow$ integrand pos.
- d) If the function f is continuous and satisfies $0 \leq f(t) \leq 1$, $t \geq 0$, then $\mathcal{L}(f)(s)$ exists for all $s > 0 \Rightarrow$ True because of $\mathcal{L}$. t. def and convergence thm.
- e) If the function f satisfies $0 \leq f(t) \leq 1$ for all $t \geq 0$, then $\mathcal{L}(f)(s)$ exists for all $s > 0 \Rightarrow$ True

Prob 4 - Initial value problems

$$F(s) = \mathcal{L}\{f'(t)\} \Rightarrow F(s) = sF(s) - f(0), \quad F(s) = \mathcal{L}\{f''(t)\} \Rightarrow F(s) = s^2 F(s) - sf'(0) - f''(0)$$

a)

$$2y'' + y' - y = 0, \quad y(0) = 0, \quad y'(0) = 1$$

$$2(s^2 Y(s) - sy(0) - y'(0)) + (sY(s) - y(0)) - Y(s) = 0$$

$$2s^2 Y(s) + sY(s) - Y(s) = 2$$

$$Y(s)(2s^2 + s - 1) = 2 \Rightarrow Y(s) = \frac{2}{(2s^2 + s - 1)} = \frac{2}{(2s-1)(s+1)}$$

Partial fraction decomposition

$$\frac{2}{(2s-1)(s+1)} = \frac{A}{(2s-1)} + \frac{B}{(s+1)} \Rightarrow 2 = s(A+2B) + A-B \Rightarrow \begin{cases} A+2B=0 \\ A-B=2 \end{cases} \Rightarrow A = \frac{4}{3}, \quad B = -\frac{2}{3}$$

$$Y(s) = \frac{\frac{4}{3}}{2s-1} - \frac{\frac{2}{3}}{s+1}$$

$$F(s) = \frac{1}{(s-a)} \Rightarrow f(t) = e^{at}$$

$$\mathcal{L}^{-1}(Y_1(s)) = \frac{4}{6} e^{\frac{1}{2}t}$$

$$\mathcal{L}^{-1}(Y_2(s)) = \frac{2}{3} e^{-t}$$

$$\underline{y(t) = \frac{4}{6} e^{\frac{1}{2}t} - \frac{2}{3} e^{-t}}$$

b)

$$y'' + y = 1, \quad y(0) = 0, \quad y'(0) = 0$$

$$(s^2 Y(s) - sy(0) - y'(0)) + Y(s) = \frac{1}{s}$$

$$Y(s)(s^2 + 1) = \frac{1}{s}$$

$$Y(s) = \frac{1}{s(s^2 + 1)}$$

Partial fraction decomposition

$$\frac{1}{s(s^2 + 1)} = \frac{A}{s} + \frac{Bs + C}{(s^2 + 1)} \Rightarrow 1 = A(s^2 + 1) + (Bs + C)s \Rightarrow \begin{cases} A+B=0 \\ C=0 \\ A=1 \end{cases} \Rightarrow A=1, B=-1, C=0$$

$$Y(s) = \frac{1}{s} - \frac{s}{(s^2 + 1)}$$

$$F(s) = \frac{1}{s} \Rightarrow f(t) = 1, \quad F(s) = \frac{s}{(s^2 + \alpha^2)} \Rightarrow f(t) = \cos(\alpha t)$$

$$\mathcal{L}^{-1}(Y_1(s)) = 1$$

$$\mathcal{L}^{-1}(Y_2(s)) = \cos(t)$$

$$\underline{y(t) = 1 - \cos(t)}$$

c) $2y'' + 10y' + 12y = 0, \quad y(0) = -2, \quad y'(0) = 1$

$$2(s^2 Y(s) - s y(0) - y'(0)) + 10(s Y(s) - y(0)) + 12 Y(s) = 0$$

$$2s^2 Y(s) + 4s - 2 + 10s Y(s) + 20 + 12 Y(s) = 0$$

$$Y(s)(2s^2 + 10s + 12) = -4s - 18$$

$$Y(s) = \frac{-4s - 18}{(2s^2 + 10s + 12)} = \frac{-4s - 18}{(2s + 6)(s + 2)}$$

Partial fraction decomposition

$$\frac{-4s - 18}{(2s + 6)(s + 2)} = \frac{A}{(2s + 6)} + \frac{B}{(s + 2)} \Rightarrow -4s - 18 = s(A + 2B) + 2A + 6B = \begin{cases} A + 2B = -4 \\ 2A + 6B = -18 \end{cases} \Rightarrow A = 6, B = -5$$

$$Y(s) = \frac{6}{(2s + 6)} - \frac{5}{(s + 2)}$$

$$F(s) = \frac{1}{(s - a)} \Rightarrow f(t) = e^{at}$$

$$\mathcal{L}^{-1}(Y_1(s)) = 3e^{-3t}$$

$$\mathcal{L}^{-1}(Y_2(s)) = 5e^{-2t}$$

$$\underline{y(t) = 3e^{-3t} - 5e^{-2t}}$$